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Protein structural phylogeny, a missing chapter in molecular evolutionary biology

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Candidate's academic degrees



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Abstract

Protein evolution has been studied intensively based on protein amino acids, i.e. the protein sequence. Thanks to the next generation sequencing, a huge number of protein sequences can be collected and compared to extract evolutionary information. The evolutionary information contributes to protein bioinformatics and structural biology fields such as protein structure prediction, protein-protein interaction prediction and phylogenetic tree analysis. When proteins evolve over long evolutionary time, their homologous relationships become remote because of the low sequence identity, which makes it difficult to explore protein functions. Thus, what is required to explain functional mechanisms is determining the high-resolution protein structure and comparatively studying superposed structures. It is because protein function is connected to protein structure in that the latter is more conserved than sequence under the evolutionary pressure. However, research on evolutionary effect of structure (i.e., structural variations) in folded protein structures has been limited, although protein structure varies during the history of life because mutations in DNA influence protein amino acid sequence and affect protein structure. Thus, I hypothesised that protein structural evolutionary information helps us to explore protein function by capturing the remote homologous relationships at the folded structure level and complement protein evolutionary information.

In my PhD project, I built a structure-based evolutionary model, developed analytic tools to reconstruct structure-based phylogenetic trees and developed a novel structure-supported pairwise sequence alignment method. Specifically, I constructed a structure-based evolutionary model based on the transition probabilities of 7 secondary structure states, laying the foundation for phylogenetic analysis of superposed structures. Furthermore, based on the concept of structural evolution, a novel pairwise sequence alignment was developed to align the sequences with a 60-state model of all combinations of 3 predicted secondary structure states and the 20 amino acids; the new alignment method outperforms the amino acid-based substitution matrix with remote homologous pairwise alignments. These techniques were further applied to superfamilies of the Toll-like/interleukin-1 receptor (TIR) domains and armadillo-repeat (ARM) domains. Members of these two superfamilies are not only difficult to analyse because of the low sequence identities but also important in biological functions. TIR domains have been studied in innate immune responses across species. My structure-based phylogenetic analysis is the first to link plant TIR domains to domains with enzyme activity, explaining signalling in plant cell death pathways. Human SARM1 (sterile alpha and TIR motif containing 1 protein) plays an important role in the Wallerian degeneration pathway, and it has an armadillo-repeat (ARM) domain, which has been hypothesised to regulate SARM1 itself. The 60-state alignment method identified a possible regulatory region, and the results have been verified using the *Drosophila* SARM1 protein.

In conclusion, this thesis proposes the first structural evolutionary model for secondary structure states, based on all high-resolution protein structures deposited in the Protein Data Bank. This model makes phylogenetic analysis possible for any superposed structures, facilitating a number of applications, including exploring protein function. Using the phylogenetic tree inference with the

novel evolutionary model and the novel pairwise alignment method developed in this PhD project, a new enzymatic function has been identified for plant TIR domains, and the binding sequence within the SARM1 protein for its ARM domain has been verified subsequently. Such critical knowledge of functional mechanisms analysed by techniques developed in this project supports the concept of the structural evolution in protein phylogenetic analysis. It is expected that structural evolutionary information for any protein structure can be studied without being limited to amino acid sequence information, revealing how protein structures evolve and bridging the gap between protein amino acid sequence and protein structure.

Declaration by author

This thesis is composed of my original work, and contains no material previously published or written by another person except where due reference has been made in the text. I have clearly stated the contribution by others to jointly-authored works that I have included in my thesis.

I have clearly stated the contribution of others to my thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial support and any other original research work used or reported in my thesis. The content of my thesis is the result of work I have carried out since the commencement of my higher degree by research candidature and does not include a substantial part of work that has been submitted to qualify for the award of any other degree or diploma in any university or other tertiary institution. I have clearly stated which parts of my thesis, if any, have been submitted to qualify for another award.

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Publications included in this thesis

1. [1] Shane Horsefield, Hayden Burdett, Xiaoxiao Zhang, Mohammad K. Manik, Yun Shi, Jian Chen, Tiancong Qi, Jonathan Gilley, **Jhih-Siang Lai**, Maxwell X. Rank, Lachlan W. Casey, Weixi Gu, Daniel J. Ericsson, Gabriel Foley, Robert O. Hughes, Todd Bosanac, Mark von Itzstein, John P. Rathjen, Jeffrey D. Nanson, Mikael Bodén, Ian B. Dry, Simon J. Williams, Brian J. Staskawicz, Michael P. Coleman, Thomas Ve, Peter N. Dodds, Bostjan Kobe, **NAD⁺ cleavage activity by animal and plant TIR domains in cell death pathways**, *Science*, 365, 6455, 2019

My contributions in this published work include a section “Bioinformatic analysis of TIR domains” of the paper.

- a) Writing specifically to the section;
- b) Data collection, annotation for the bioinformatics analysis;
- c) Preparation of figures in the section and the supplementary information;
- d) Designing scientific concept;

The section “Bioinformatic analysis of TIR domains” of this published work was included in the Chapter 4 of the thesis.

2. [2] **Jhih-Siang Lai**, Burkhard Rost, Bostjan Kobe, Mikael Bodén,, **Evolutionary model of protein secondary structure capable of revealing new biological relationships**, *Proteins: Structure, Function, and Bioinformatics*, 2020, <https://doi.org/10.1002/prot.25898>

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- a) Writing for the manuscript;
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